

EXAM 7 – SPRING 2014

19. (2.25 points)

A U.S.-domiciled insurance company is holding the following bonds:

- Bonds issued by a U.S.-based bank with credit rating of AA valued at \$15 million. The mean time to maturity is five years.
- Bonds issued by a technology company located in the U.S. with credit rating of BBB valued at \$20 million. The mean time to maturity is 10 years.

There are no other bonds held by the company.

Identify and briefly describe three key drivers of credit risk for the insurance company and briefly comment on how they relate to the insurer's bond portfolio.

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SAMPLE ANSWERS AND EXAMINER'S REPORT

QUESTION 19	
TOTAL POINT VALUE: 2.25	LEARNING OBJECTIVE: C1: Demonstrate how insurance and financial risk can be analyzed quantitatively
SAMPLE ANSWERS	
<u>Credit quality</u> (also accepted counterparty risk, default risk, and credit rating)	
<i>Sample Answer 1</i>	
Credit quality of an investment or an enterprise refers to the probability that the issuer will meet all contractual obligations. This assessment normally occurs at both the initial investment and at each renewal point. One of the common measurements used in assessing credit quality is the rating assigned to the issuer. A variety of ratings agencies provide these assessments to the public, giving the investor a perceived level of confidence in the issuer's ability to make good on the repayment schedules to which it is committed.	
<i>Sample Answer 2</i>	
The credit risk from credit quality is higher for the technology company bonds according to the credit ratings.	
<u>Maturity</u> (also accepted time horizon or duration)	
<i>Sample Answer 1</i>	
The longer the term to maturity of an investment, the longer even a high quality issuer has to potentially deteriorate.	
<i>Sample Answer 2</i>	
The credit risk from maturity is higher for the bank bonds since they have a longer maturity.	
<u>Concentration by industry</u> (also accepted concentration by type)	
<i>Sample Answer 1</i>	
Conditions that trigger credit events have a tendency to impact on the entire economy simultaneously. Within this general characteristic, however, the impact of economic development often varies between sectors of the economy. Within a sector, however,	

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there tends to be uniformity between the entities participating in that sector. Degrees of separation within a sector will exist, but these are on a smaller scale than those that normally occur between sectors.

Sample Answer 2

The credit risk from concentration by industry is reduced by choosing bonds from companies in two unrelated industries.

Concentration by geography (also accepted concentration by region/country)

Sample Answer 1

Credit risk has been shown to carry a large degree of contagion. Periods of relatively few credit events are followed by periods where default experience is extremely high. Similarly, economically depressed regions tend to produce high levels of default experience in comparison with more prosperous areas. That these regions can and do change over time creates a challenge to the process of credit risk analysis.

Sample Answer 2

There is some credit risk from concentration by geography since all of the bonds are from American companies. OR The credit risk from concentration by geography is reduced by investing in bonds from a national rather than regional bank.

Size of expected loss

Sample Answer 1

The size of loss due to a credit event can vary widely, from loss of some or all of the return on an investment to loss of some, or all, of the inherent principal. Losses can also occur from a delay in the timing of a scheduled payment, causing either a loss of return during the deferral period, a reduction in available reinvestment rate during the deferral period, or both. When a scheduled payment is delayed for any reason, there is also the potential for an associated loss if the payment were needed to match a scheduled outflow. The investor would then be required to make good on its obligation by borrowing or selling other assets. They might need to delay payment of their own scheduled obligation, possibly incurring a penalty.

SAMPLE ANSWERS AND EXAMINER'S REPORT

Sample Answer 2

There is not enough information to determine the amount of credit risk from size of expected loss.

EXAMINER'S REPORT

- Candidates were expected to identify, briefly describe, and comment on the application of three key drivers of credit risk to the bond portfolio.
- Most candidates were successful in identifying, describing, and commenting on at least 2 key drivers. Many candidates successfully identified and commented on 3 key drivers, with the descriptions of the drivers being absent or insufficient.
- Additional common responses and remarks:
 - Downgrade risk was a very common incorrect response. Partial credit was given when the response included an appropriate discussion of maturity.
 - While not explicitly identified in the reading, credit was granted for responses describing lack of diversification with respect to the number of issuers.
 - Some candidates listed sovereign risk as a key driver, and this was considered an incorrect response. However, partial credit was given if this response included an appropriate discussion relating to being concentrated in the US.
 - Settlement risk was another incorrect response given by some candidates. Partial credit was given when the response included an appropriate discussion of maturity. Payment lag was not acceptable. Discussions of interest rate or liquidity risk were not accepted since this is a separate issue from credit risk.
 - Descriptions and comments for “size of expected loss” pertaining to the size of the bond values were not accepted, as this pertains more to portfolio concentration risk with respect to lack of diversification.